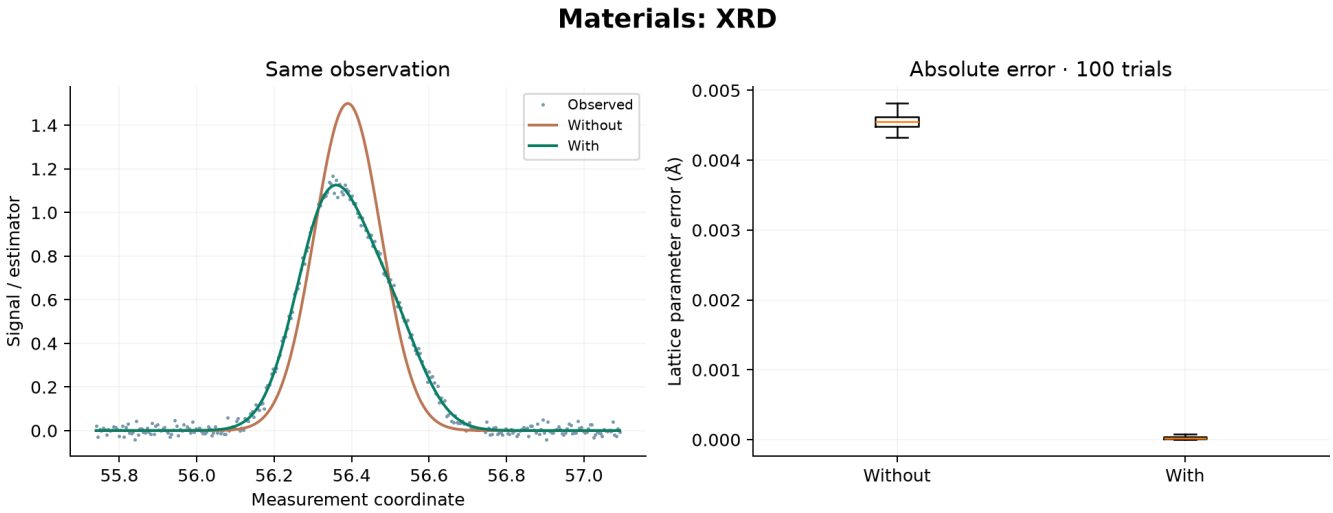
Cross-domain analysis: supplementary results

Paired synthetic pilot, 100 trials per field. Without and With use the same observation; With adds a domain-specific physical constraint or reference correction. Improvements are reductions in mean absolute parameter error, not classification accuracy or real-world validation.

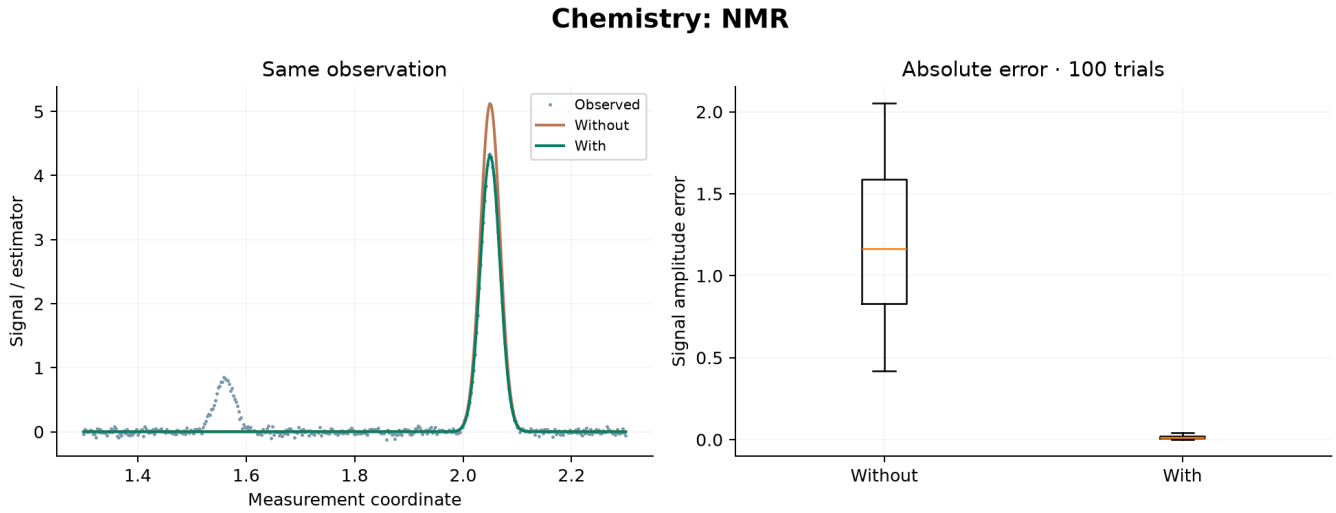
Materials: XRD



Mean error reduction: +99.4% (paired bootstrap 95% interval +99.3% to +99.5%; n=100).

Synthetic proof of concept. This does not establish held-out performance of the full RamanLens system.

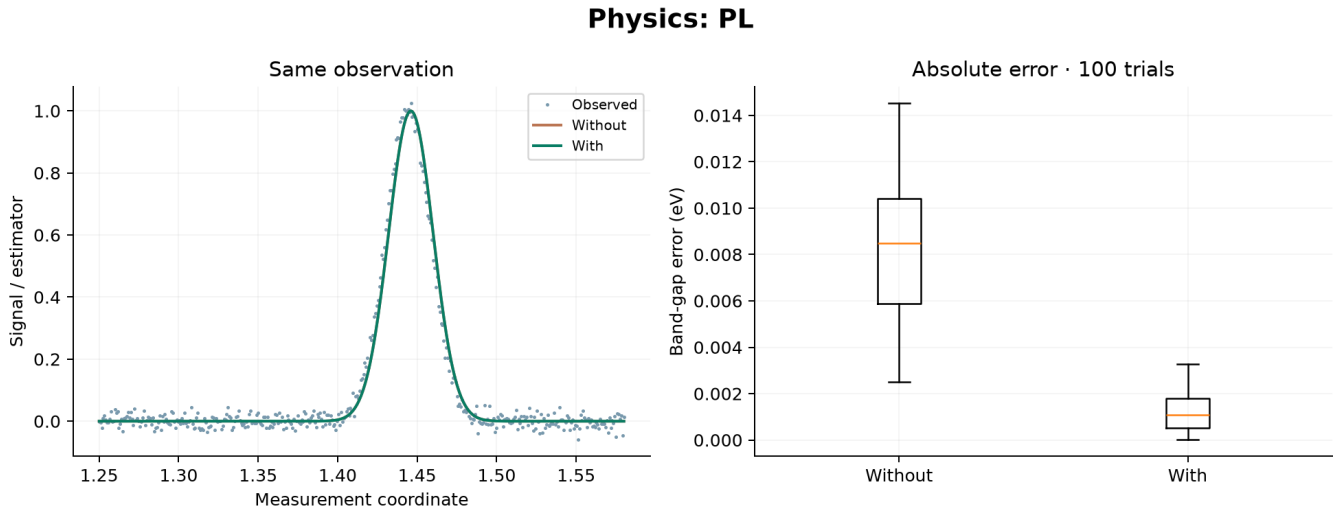
Chemistry: NMR



Mean error reduction: +98.8% (paired bootstrap 95% interval +98.6% to +98.9%; n=100).

Synthetic proof of concept. This does not establish held-out performance of the full RamanLens system.

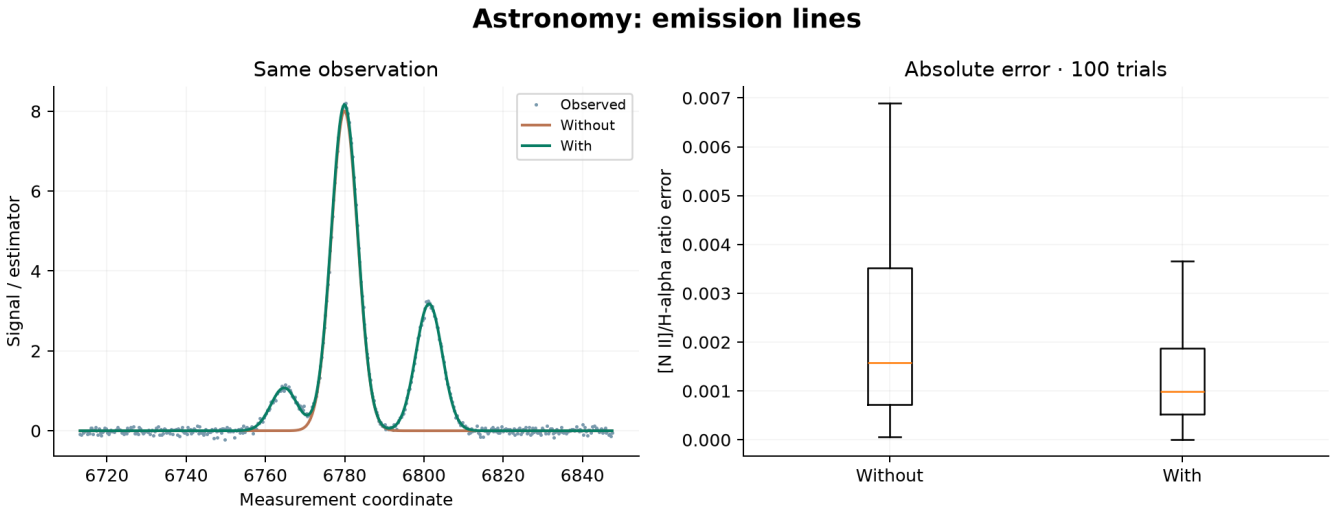
Physics: PL



Mean error reduction: +84.7% (paired bootstrap 95% interval +82.5% to +86.8%; n=100).

Synthetic proof of concept. This does not establish held-out performance of the full RamanLens system.

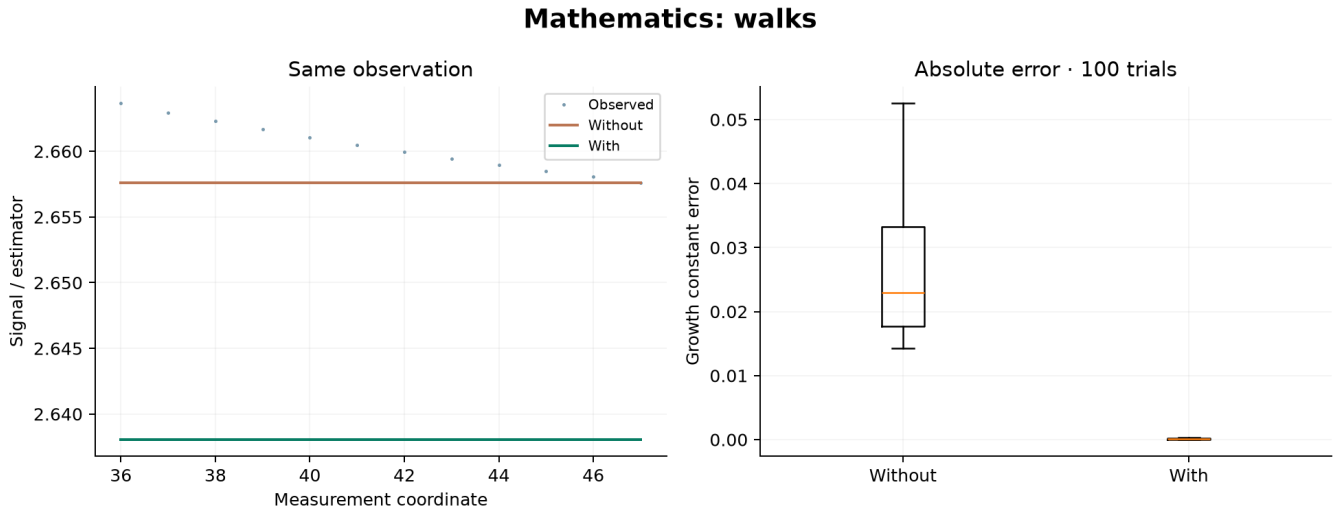
Astronomy: emission lines



Mean error reduction: +42.0% (paired bootstrap 95% interval +27.3% to +53.5%; n=100).

Synthetic proof of concept. This does not establish held-out performance of the full RamanLens system.

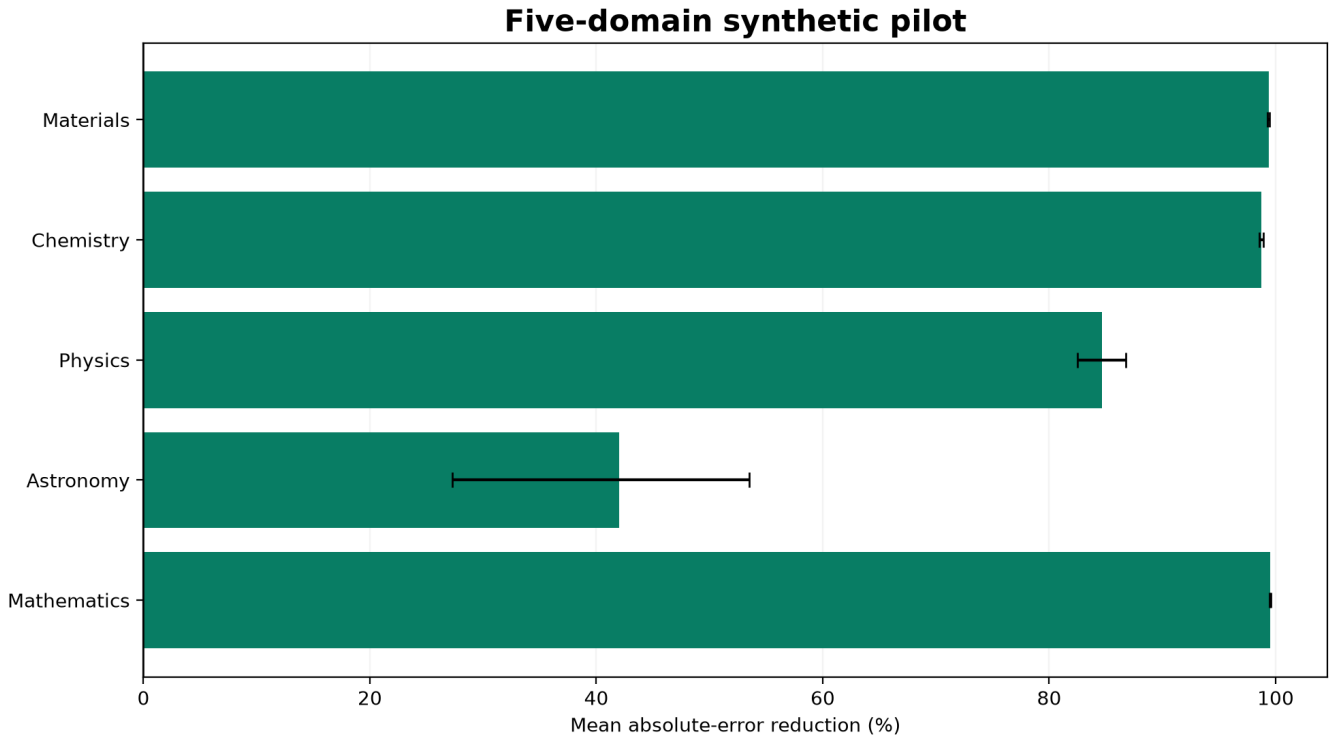
Mathematics: walks



Mean error reduction: +99.5% (paired bootstrap 95% interval +99.4% to +99.6%; n=100).

Synthetic proof of concept. This does not establish held-out performance of the full RamanLens system.

Summary



All five results are paired synthetic method comparisons. They use different scientific quantities and deliberately simplified models. The published Raman held-out benchmark must be reported separately rather than pooled with these five error-reduction percentages.